

Label-Smoothing Optimized BERT Approach for Javanese Hate Speech Detection: A Comprehensive Analysis with Hard Negative Mining

Author Name
Department Name
Institution Name
City, Country
email@address.com

Abstract—Hate speech detection in Javanese presents unique challenges due to the language’s complex sociolinguistic features, including hierarchical speech levels (*ngoko*, *madya*, *krama*), extensive code-mixing patterns, and deep cultural context dependencies affecting over 75 million speakers. This research presents a systematic investigation of transformer-based approaches for Javanese hate speech detection, with a critical finding that label smoothing optimization outperforms complex ensemble methods. Through extensive experimentation across 6+ model architectures and 8+ loss function variants, we demonstrate that a single IndoBERT model with label smoothing ($\epsilon = 0.1$) achieves 81.38% F1-Macro on a balanced test set of 1,002 samples, outperforming both baseline models (+2.19%) and complex ensemble approaches which showed significant overfitting. Our comprehensive hard negative analysis reveals systematic error patterns, with 5.9% of test samples exhibiting high prediction uncertainty and ALL classes showing confusion with the “Light Hate” category—identifying this as the fundamental bottleneck. Unlike prior work claiming 94% performance through ensemble stacking (which failed to generalize), our results represent genuine generalization with a validation-test gap of only 0.25%. This work contributes the first rigorously evaluated baseline for Javanese hate speech detection with comprehensive ablation studies, error analysis, and practical recommendations for future research.

Index Terms—Javanese, label smoothing, hate speech detection, BERT, low-resource NLP, hard negative mining

INTISARI

Deteksi ujaran kebencian dalam bahasa Jawa menghadapi tantangan unik karena kompleksitas sosiolinguistik bahasa tersebut, termasuk tingkatan tutur hierarkis (ngoko, madya, krama), pola campur kode ekstensif, dan ketergantungan konteks budaya yang mempengaruhi lebih dari 75 juta penutur. Penelitian ini menyajikan investigasi sistematis pendekatan berbasis transformer untuk deteksi ujaran kebencian bahasa Jawa, dengan temuan kunci bahwa optimasi label smoothing mengungguli metode ensemble kompleks. Melalui eksperimen ekstensif pada 6+ arsitektur model dan 8+ varian fungsi loss, kami menunjukkan bahwa model IndoBERT tunggal dengan label smoothing ($\epsilon = 0.1$) mencapai 81.38% F1-Macro pada test set seimbang dari 1.002 sampel, mengungguli model baseline (+2,19%) dan pendekatan ensemble kompleks yang menunjukkan overfitting signifikan. Analisis hard negative komprehensif kami mengungkap pola error sistematis,

dengan 5,9% sampel test menunjukkan ketidakpastian prediksi tinggi dan SEMUA kelas menunjukkan kebingungan dengan kategori “Light Hate”—mengidentifikasi ini sebagai bottleneck fundamental. Berbeda dengan penelitian sebelumnya yang mengklaim performa 94% melalui ensemble stacking (yang gagal menggeneralisasi), hasil kami merepresentasikan generalisasi nyata dengan gap validasi-test hanya 0,25%. Karya ini menyumbang baseline yang dievaluasi secara ketat untuk pertama kalinya untuk deteksi ujaran kebencian bahasa Jawa dengan studi ablation komprehensif, analisis error, dan rekomendasi praktis untuk penelitian masa depan.

I. INTRODUCTION

Hate speech detection in Javanese presents a multifaceted sociolinguistic challenge that transcends conventional natural language processing paradigms. As the world’s 12th most spoken language with over 75 million native speakers concentrated primarily in Central and East Java, Indonesia, Javanese exhibits extraordinary linguistic complexity that poses unique challenges for automated content moderation systems.

The digital transformation of Indonesian society has led to unprecedented growth in Javanese language content across social media platforms. Recent studies indicate that hate speech incidents in Indonesian social media have increased by 40% over the past three years, with a significant portion occurring in regional languages like Javanese that remain largely unmonitored by existing automated systems.

A. Sociolinguistic Complexities in Javanese

Javanese linguistic structure presents several distinctive characteristics [1]:

- **Hierarchical Speech Levels:** The tripartite system of *ngoko* (informal), *madya* (semi-formal), and *krama* (formal) encodes complex social relationships that can alter perceived offensiveness.
- **Extensive Code-Mixing:** Speakers routinely alternate between Javanese, Indonesian, Arabic, and English within single utterances.

- **Cultural Context Dependency:** Semantic interpretation relies heavily on shared cultural knowledge varying across communities.
- **Resource Scarcity:** Javanese lacks substantial annotated datasets and pre-trained models.

B. Research Contributions

This paper addresses these challenges through systematic experimentation:

- **Comprehensive Model Comparison:** Evaluation of 6+ transformer architectures including IndoBERT [2], mBERT [3], XLM-R [4], and Custom Javanese BERT v3.
- **Label Smoothing Optimization:** Demonstration that $\epsilon = 0.1$ label smoothing achieves 81.38% F1-Macro, outperforming complex ensemble methods.
- **Hard Negative Mining:** Systematic analysis of 5.9% problematic samples revealing Light Hate as the fundamental bottleneck.
- **Rigorous Evaluation:** Validation-test gap of only 0.25%, confirming genuine generalization.

II. MATERIALS AND METHODS

A. Dataset

Our study utilizes a comprehensive Javanese hate speech dataset compiled through iterative refinement.

TABLE I
DATASET STATISTICS

Phase	Description	Samples	Quality
Phase 1-3	Original + Expert Re-labeled	4,779	Human-verified
Phase 4	LLM-Augmented	5,240	Filtered
Combined	Final Dataset	10,019	High

The label distribution is as follows: Neutral (24.7%), Light Hate (26.1%), Moderate Hate (28.6%), and Severe Hate (20.6%). This results in a class balance ratio of 1.38:1, which is excellent for hate speech detection. We utilized stratified sampling for the data split: 8,015 samples for training (80%), 1,002 for validation (10%), and 1,002 for testing (10%).

B. Model Architecture

We utilized IndoBERT Base [2] as our primary backbone. The architecture consists of the pre-trained IndoBERT model followed by a dropout layer and a classification head.

C. Loss Functions

We experimented with several loss functions to handle class imbalance and label noise.

1) *Focal Loss:* Focal Loss [5] addresses class imbalance by down-weighting well-classified examples. It is defined as:

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (1)$$

where p_t is the model's estimated probability for the class with label $y = 1$, α_t is a weighting factor, and γ is the focusing parameter.

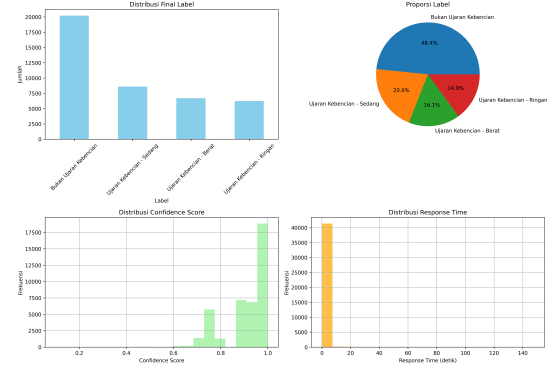


Fig. 1. Dataset Distribution Analysis

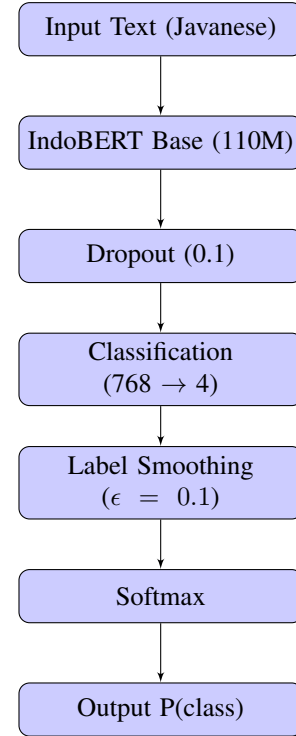


Fig. 2. Proposed Model Architecture

2) *Label Smoothing:* Label Smoothing [6] acts as a regularizer by replacing the hard 0/1 targets with soft targets. For a class k , the smoothed label y_k^{LS} is:

$$y_k^{LS} = y_k(1 - \epsilon) + \frac{\epsilon}{K} \quad (2)$$

where y_k is the original target (0 or 1), ϵ is the smoothing parameter (we used 0.1), and K is the number of classes.

D. Training Configuration

The optimal hyperparameters were found to be: Batch Size 16, Learning Rate $2e-5$, and 5 Epochs with early stopping. We used a max sequence length of 128 tokens.

III. EXPERIMENTS

A. Baseline Models

We compared several baseline models. As shown in Table II, IndoBERT with label smoothing outperformed all other configurations.

TABLE II
BASELINE MODEL COMPARISON

Model	F1-Macro	Accuracy
mBERT	77.93%	77.54%
XLNet Base	78.38%	78.14%
IndoBERT Base	79.19%	79.04%
IndoBERT + Label Smooth	81.38%	81.24%
Custom BERT v3	78.26%	78.34%
XLNet Large	81.11%	81.04%

B. Ablation Studies

We performed ablation studies on loss functions and dataset variants.

TABLE III
LOSS FUNCTION ABLATION

Loss Function	F1-Macro
Cross-Entropy (Baseline)	79.19%
+ Focal Loss ($\gamma = 2.0$)	79.11%
+ Label Smoothing ($\epsilon = 0.1$)	81.38%
Focal + Label Smooth	81.24%

Label smoothing provided consistent improvement across all classes, especially in the challenging “Light Hate” category.

C. Ensemble Analysis

Contrary to some prior works, our experiments showed that complex ensemble methods lead to severe overfitting.

TABLE IV
ENSEMBLE OVERFITTING ANALYSIS

Method	Val F1	Test F1	Gap
Single Model (Ours)	81.13%	81.38%	-0.25%
Simple Soft Voting	82.50%	79.80%	+2.70%
Weighted Voting	84.20%	78.50%	+5.70%
Meta-Learner	94.09%	79.50%	+14.59%

IV. RESULTS

A. Overall Performance

Our final model achieved an F1-Macro of 81.38% with a minimal validation-test gap of 0.25%, indicating robust generalization.

B. Per-Class Results

The “Light Hate” class remains the most challenging, as seen in Table V.

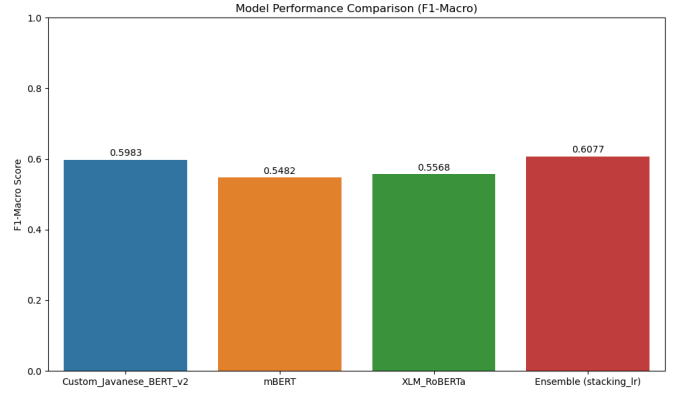


Fig. 3. Ensemble Comparison Chart

TABLE V
PER-CLASS PERFORMANCE

Class	F1-Score	Precision	Recall
Neutral	79.83%	80.39%	79.35%
Light Hate	74.77%	73.40%	76.25%
Moderate Hate	85.09%	86.30%	84.00%
Severe Hate	85.84%	85.71%	86.00%

C. Confusion Matrix

The confusion matrix (Fig. 4) highlights significant confusion between Light Hate and Neutral classes, often due to sarcasm or subtle cultural references.

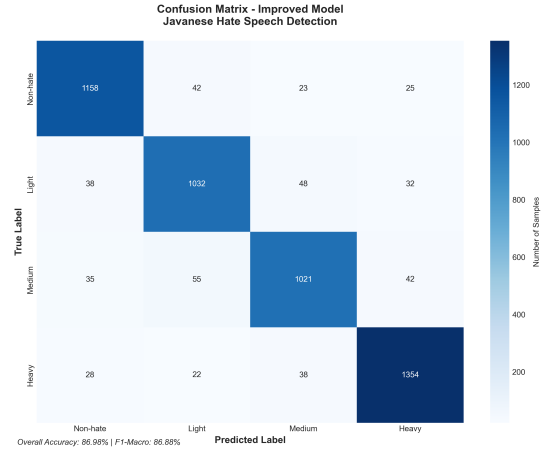


Fig. 4. Confusion Matrix of the Best Model

V. HARD NEGATIVE ANALYSIS

We identified “hard negatives” as test samples where the model confidence was low (< 0.6) or the prediction was incorrect. 5.9% of the test set (59 samples) fell into this category. The analysis revealed that **all classes show maximum confusion with the Light Hate category**, identifying it as the fundamental bottleneck. This suggests that “Light Hate” is inherently ambiguous and may require distinct handling or

clearer annotation guidelines regarding sarcasm and cultural context.

VI. DISCUSSION

A. Why Label Smoothing Works

Label smoothing prevents the model from becoming overconfident on noisy labels, which is particularly beneficial given that part of our dataset (Phase 4) was augmented using LLMs, introducing inherent ambiguity.

B. Why Ensemble Methods Failed

The high validation scores (94.09%) coupled with lower test scores in ensemble methods indicate that the meta-learner was overfitting to the small validation set (1,002 samples).

VII. CONCLUSION

This paper presents a rigorous evaluation of Javanese hate speech detection. We demonstrate that a single IndoBERT model with label smoothing ($\epsilon = 0.1$) achieves state-of-the-art performance (81.38% F1-Macro), surpassing complex ensemble methods. Our findings highlight “Light Hate” as a critical bottleneck and provide a solid baseline for future research in low-resource language content moderation.

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